

| Peripherals                        |                | STM32C542FCP                              | STM32C542FCU | STM32C542ECU            | STM32C542KCT            | STM32C542KCU | STM32C542CCT<br>STM32C542CCU | STM32C542RCT             |
|------------------------------------|----------------|-------------------------------------------|--------------|-------------------------|-------------------------|--------------|------------------------------|--------------------------|
| Timers                             | Low-power      | 1                                         |              |                         |                         |              |                              |                          |
|                                    | SysTick        | 1                                         |              |                         |                         |              |                              |                          |
|                                    | Watchdog       | 2                                         |              |                         |                         |              |                              |                          |
| Communication<br>interfaces        | SPI (with I2S) | 2                                         |              |                         |                         |              |                              |                          |
|                                    | I2C            | 1                                         |              |                         |                         |              |                              |                          |
|                                    | I3C            | 1                                         |              |                         |                         |              |                              |                          |
|                                    | USART          | 2                                         |              |                         |                         |              |                              |                          |
|                                    | UART           | 1                                         |              |                         | 2                       |              |                              |                          |
|                                    | LPUART         | 1                                         |              |                         |                         |              |                              |                          |
|                                    | USB            | 1                                         |              |                         |                         |              |                              |                          |
|                                    | FDCAN          | 2                                         |              |                         |                         |              |                              |                          |
|                                    | TRACE          | No                                        |              |                         |                         |              |                              | Yes                      |
| CORDIC                             |                | Yes                                       |              |                         |                         |              |                              |                          |
| RTC                                |                | Yes (without LSE)                         |              |                         |                         |              | Yes                          |                          |
| Tamper pins                        |                | 1                                         |              | 2                       |                         |              | 3                            |                          |
| True random number generator (RNG) |                | Yes                                       |              |                         |                         |              |                              |                          |
| HASH                               |                | Yes                                       |              |                         |                         |              |                              |                          |
| AES                                |                | Yes                                       |              |                         |                         |              |                              |                          |
| GPIOs                              |                | 14                                        |              | 16                      | 25                      | 27           | 38                           | 52                       |
| Wake-up pins                       |                | 1                                         |              | 3                       |                         |              | 4                            |                          |
| 12-bit ADC channels                |                | 4 external + 2 internal                   |              | 5 external + 2 internal | 8 external + 2 internal |              |                              | 12 external + 2 internal |
| 12-bit DAC channels                |                | 2                                         |              |                         |                         |              |                              |                          |
| Analog comparator                  |                | 2                                         |              |                         |                         |              |                              |                          |
| Operational amplifier              |                | 1                                         |              |                         |                         |              |                              |                          |
| Maximum CPU frequency (MHz)        |                | 144                                       |              |                         |                         |              |                              |                          |
| Operating voltage                  |                | 2.7 V - 3.6 V                             |              |                         |                         |              |                              |                          |
| Operating temperature              |                | Junction temperature range: -40 to +140°C |              |                         |                         |              |                              |                          |
| Packages                           |                | TSSOP20                                   | UFQFPN20     | UFQFPN24                | LQFP32                  | UFQFPN32     | LQFP48<br>UFQFPN48           | LQFP64                   |

Figure 1 shows the general block diagram of the device family.